DOCUMENT TITLE	REVISION
SRP-043 – Front Suspension – Preliminary Design Report	0

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Department	Project
Vehicle Dynamics	Front Suspension

REVISION HISTORY

Rev.	Change Log	Author/s	Date
0	Created Report	James Crisp	17/09/2024

1. Key Components

1.1 Key Component Definition

SR8 will use a double wishbone setup in the front and rear with a direct coil over. Table 1.1 categorises the key components that are fundamental to the front suspension assembly

Table 1: Key Components

Project:	Front Suspension
Key Components:	- Upper control arm: used to align the wheel and defines its motion and geometry across its length of travel. - Lower control arm - Chassis Brackets: links the control arms to the chassis - Upright Brackets: connects the upright to the control arms - Damper: absorbs and damps the oscillation of the coil - Coil: absorbs oscillations from the wheel

Project:	Wishbone
Key Components:	- Carbon fibre linkage: structural member of the control arm. Their lengths define the suspension geometry - Rod ends: spherical bearings for reliable motion of control arms - Aluminium inserts: Threaded insert for rod ends to be fastened to the linkage

Commented [DR1]: A little sentence or phrase on what each of these key components are would be good

2. Design Factors

2.1 Lifecycle Analysis

The suspension system of the car should last the lifetime of the vehicle, with certain serviceable components being replaced at required intervals. Components expected to last the lifetime of the vehicle include the wishbones, chassis mounts and coil. Serviceable components include dampers and rod ends. Rod ends should be inspected before and after each race to assess any wear, with more in-depth services taking place every 1000 hours of operation.

Commented [DR2]: Do we know roughly what these regular intervals are?

Materials used in the suspension assembly include carbon fibre for the linkages, aluminium for inserts at the end of each linkage, and steel rod ends. Since the assembly is expected to last the lifetime of the vehicle on the end of its life it will likely be scrapped as carbon fibre is not easily recyclable and the inserts are pressed fit and epoxied in place. If the steel rod ends are in good condition they could be sold or recycled.

2.2 Risk Analysis

Major concerns related to suspension are related to the failure of critical components such as control arms, pushrods, dampers, springs etc. There is also the gradual failure of ball joints due to wear. This gradual wear can result in a car that does not respond correctly to driver inputs due to excessive play in joints. A full list of identified risks is found in Table 2 below

Commented [DR3]: This part is very good. But also are there any risks in the manufacture and assembly stages? Risks to the students working with them / installing them?

Commented [DR3R2]: Maybe also include the risk associated with corrosion based on today's workshop.

Table 2: Risk Analysis Table

Risk	Response Category	Mitigation Strategy
Suspension Hardware		
Ball joint wear	Accept	Regular servicing/inspections which are included as a vehicle inspection check. See lifecycle for lifespan of these components.
Bushing Wear	Accept	Regular servicing/inspections which are included as a vehicle inspection check. See lifecycle for lifespan of these components.
Damper leak	Mitigate	Replace/Service the faulty equipment when leak identified
Installation Risks		
Misalignment	Mitigate	Small deviations in pickup point locations should be measured. These misalignments will be almost inevitable in installation and the manufacture of the chassis but can be moderately counteracted by varying each linkages length.
Coil Spring Compression	Mitigate	When installing or uninstalling the coil needs to be compressed so it fits. A compressed spring is extremely dangerous if handled without care. To mitigate this spring compressors should be used, and the installer should have a clear plan of what to once the spring is compressed the minimise time between compression and installation.
Composite Manufacturing Issues	Mitigate	The linkages are a key structural component Attend Aditya's' composite workshop and work through the native Lab composite manufacturing modules to avoid air entrapment, delamination, uneven resin distribution and other composite manufacturing risks. Having the

		supervision of a SunSwift member with composite experience will alleviate these risks also.
Critical Failure Modes		
Composite spring failure	Mitigate	Regular inspections which are included as vehicle inspection check. Replace spring if any damage to material
Control arm failure	Mitigate	Regular inspections which are included as vehicle inspection check. Replace control arm if any damage to material
Loose hardware	Mitigate	Regular inspections which are included as vehicle inspection check. Correct use of fasteners and thread locking compounds.
Project Unknowns		
Composite spring unavailable in time	Avoid	The car will be able to use both conventional coil overs or a transverse leaf spring to account for the experimental nature of the leaf spring development.

2.3 Innovation Analysis

Commented [DR4]: This is very good and thorough.

The suspension geometry itself has been thoroughly studied over the years, leaving little room for significant innovation. Systems like the Double Wishbone, commonly found in high-performance vehicles, offer minimal innovation but also very low risk due to their proven reliability. Therefore, while the geometry remains unchanged, material choices still present some opportunity for development. The decision on materials will be made once the forces acting on the structural members are fully understood, with carbon fibre being a likely candidate.

However, there are several other components of the suspension assembly—such as dampers, springs, and joints—that offer more room for innovation.

One promising area of innovation is in the use of flexure joints for the control arms. Unlike spherical bearings, flexure joints are lighter, consist of fewer parts, and eliminate the sliding friction losses found in traditional designs. These characteristics make them more reliable, predictable, and consistent under load. These joints, which can be made from carbon fibre or metallic alloys, would need to be checked frequently for cracks and other signs of wear as their operation is critical to the safety of the driver. Flexure joints could be a valuable focus for research, particularly for someone with a background in material science and composite materials looking for a thesis project. They can also be retrofitted to the SR8 with a redesigned control arm that aligns with existing pickup points.

Given their medium risk but high potential for innovation, they are worth serious consideration for future development.

Another intriguing concept is the composite transverse leaf spring, which is relatively new to the automotive industry. This component combines the functions of both the spring and anti-roll bar into a single unit. It offers excellent spring rate performance while significantly reducing weight compared to traditional systems. If designed correctly—ensuring that internal micro-strain levels remain below critical thresholds—the composite leaf spring could provide a maintenance-free, high-performance solution for the SR8's suspension needs.

However, manufacturing such a spring would require highly specialized expertise and custom dimensions tailored to the SR8, which would likely necessitate external collaboration with experts like Advance Composite Structure Australia (ACSA) and a thesis student. While this option is high in innovation, it also presents a high risk, primarily due to potential delays and the associated cost. Since no current thesis students are working on this system, it is unlikely to be completed in time for the SR8 assembly, making it impractical for immediate consideration.

Diresh's thesis work is focused on programming an active damper system for the SR8. These dampers would adjust in real-time to varying road conditions, optimizing the damping ratio for different terrains and driving conditions. On rough roads or twisty circuits, this system could greatly enhance performance by maximizing the tire's contact with the road surface. Active dampers represent a compromise-free solution, ensuring the suspension is suited to both high and low speeds, as well as long straights or tight corners. They offer medium risk but a high degree of innovation, making them a viable solution for the SR8.

Finally, a regenerative damper system has been explored for the SR8. This system would use the same mounting points as conventional dampers but, instead of dissipating mechanical energy as waste heat, it would convert it into electricity, which could be fed back into the battery. While this concept shows promise, particularly on rougher surfaces where energy recovery is more effective, its usefulness on smooth roads—where the SR8 is primarily intended to operate—is limited. Although the regenerative damper is highly innovative, the high risk of implementation and its limited benefits for the SR8 make it less practical for this project.

Table 3: Innovation Analysis

Low gains *Achievable* *Impractical*

Component	Innovation	Risk	Overall
Bushings	Low	Low	
Ball-Joints	Low	Low	
Transverse leaf spring	High	High	
Regenerative dampers	High	High	
Double wishbone	Med	Low	
Flexure joints	Med	Med	
Active Damper	High	Med	

Commented [DR5]: Could we add a key or words for what the colours represent please?

2.4 Manufacture and Procurement

Commented [DR6]: It would be worth to have a small brief on how we want to manufacture the in house ones and the materials they will be made of.

Commented [DR6R2]: Doesn't have to be that in depth at this stage

Manufacture of the suspension system would be primarily in-house, with certain components where there would be little to no benefit from attempting to produce the component in-house.

Table 4: Manufacture and Procurement

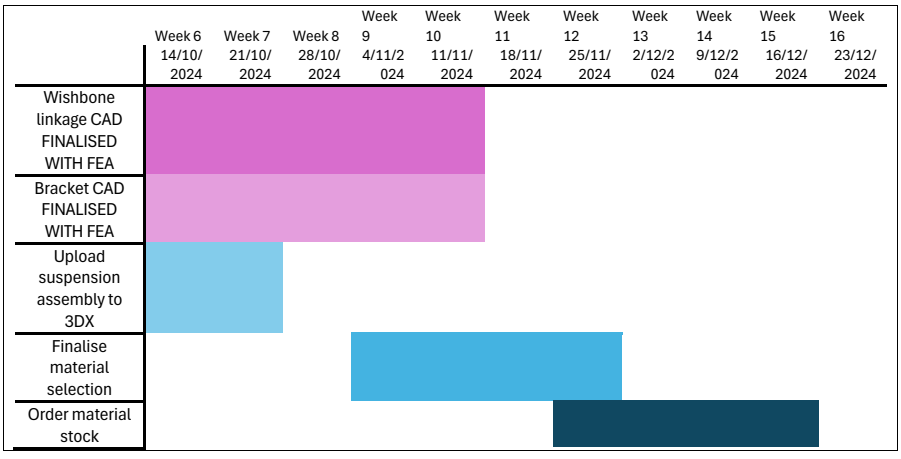
Subsystem	Method
Control Arms	Manufacture in house/Sorensen: Epoxy aluminium inserts and linkages together
Female Rod Ends	Source from SKF
Aluminium Inserts	Manufacture in house: Utilise a lathe to thread and face aluminium
Aluminium upright bracket	Manufacture in house: CNC will be required. Design MUST take this into account and be reviewed to ensure geometry is manufacturable.
Steel chassis brackets	Manufacture in house: Weld steel plates like SR7
Carbon Fibre Linkages	Manufacture in house: wet layup carbon fibre tubes
Dampers	Bilstein Suspension

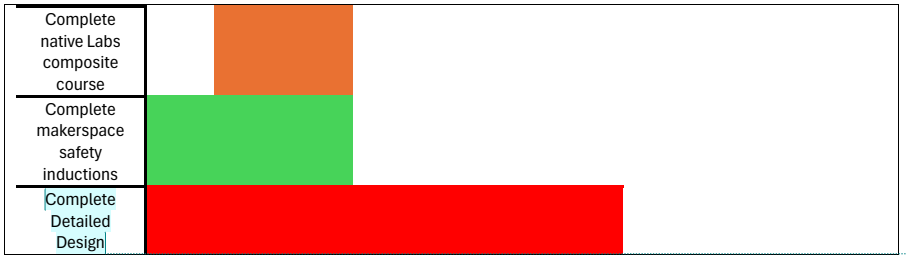
2.5 Design and Production Plan

Remaining key milestones include:

- Detailed Design
- Construction

The detailed design timeline has been broken down below:





Commented [AD7]: Prelim? - vai

2.6 Cost Analysis

A cost analysis is fundamental and will require a more detailed design to calculate a better estimate. A conceptual estimate is provided below

Component	Cost per unit	Approx Qty	Total Cost
Female rod ends	$270[1]	16	$4320
Spherical bearings	$141[2]	8	$1128
Active Damper	High	4	-
Carbon fibre	$70[3] /m^2	16 metres	$1120
Aluminium	$582[4]	150 x 150 x 700 mm	$582
			$7150

Commented [DR8]: Maybe just add another column for the total cost (cost per unit x approx qty)

1. https://appliedau.com.au/product-details/a1b6027d-d7c0-11ea-9e6c-567e73c68856 is the rod-end we are currently going for. More detailed lifecycle analysis needed to be conducted for the detailed design
2. https://appliedau.com.au/product-details/9e55263a-d7c0-11ea-9e6c-567e73c68856 press fit into the upright bracket for lower and upper control arms
3. https://trojanfibreglass.com.au/product/carbon-fibre-double-bias-450gm-1200mm/ (taken from a Sydney supplier as an estimate, unsure what supplier SR8 will use
4. Quote from Action Aluminium in western Sydney with preliminary estimates of stock needs from SolidWorks CAM

2.7 Educational Outcome Analysis

If manufacture were to be viable and safety is a factor that can be upheld, the process of manufacture would provide an exceptional learning opportunity for Vehicle Dynamics and more broadly the Mechanical team.

This learning experience would require communications with McConaughy and potentially Sorensen to ensure that safety items are met.

The students will be able to:

- Design and optimise a suspension geometry
- Research off-the-shelf components to satisfy the functional requirements
- Generate CAD and CAM models as well as engineering drawings for the manufacture of components
- Perform FEA analysis on critical components
- Gain experience with composite manufacturing and on metal working machines such as a lathe and milling machine
- Collaborate efficiently with other departments such as Chassis & Bodywork, and powertrain

2.8 Upskilling Development

While the manufacture of certain components may be outsourced but, inductions in Mill and Lathe would help the team understand the limitations of manufacturing methods, so that they can adjust their designs accordingly. Welding could also prove to be useful depending on the chosen manufacturing methods.

The OptimumG software is a powerful tool that is used heavily in the design of a suspension model. The team must be trained on this software to ensure it is used to its full potential. There are online tutorials available online from OptimumG that clearly explain the software.

- OptimumG Series
  - OptimumDynamics
  - OptimumKinematics

Additional upskilling for the manufacturing and design of components includes:

- ANSYS FEA
- Native Lab composite manufacturing courses
- Machining induction at the Makerspace

3. Requirements

3.1 Australian Design Rules ADRs

Table 5: ADR requirements

Reference	Title	Description	TPM
REQ4.1	ADR43 Section 6.1.4.1.1	Complies with minimum loaded ride height	TPM 401 – for the mid-point between any 2 consecutive '*Axles*', the dimension in millimetres obtained by multiplying the distance between those 2 '*Axles*' in metres by 33.33
REQ4.2	ADR43 Section 6.1.5.1	Complies with overall vehicle width	TPM 402 – Total vehicle width must not exceed 2,500 mm
REQ4.3	ADR43 Section 6.2.6.2	Complies with minimum unladen ride height	TPM 403 – 'Running' Clearance must not be less than 100 mm.
REQ4.4	ADR43 Section 8.1 8.1.1.	All the 'Axles' in an 'Axle Group' other than a	

Commented [DR9]: Need some more requirements for the emergency scenarios and user error -> what must happen if certain components fail?

Any safety requirements?

Reliability requirements? - i.e. how long must certain components last? Other reliability factors?

Commented [JC9R2]: Maintenance has been covered in normal operations (REQ1.8) and safety has also been covered in normal operations (REQ1.7 and REQ1.9). Safety factors can be calculated once bolts and rod-ends have been selected but are usually close to 1 and over spec-ing rod ends leads to large increases in cost, weight and size.

		'Twin Steer Axle Group' must be related to each other through a 'Load Sharing Suspension' except: 8.1.1. this requirement must not apply to a 'Close Coupled Axle Group' on any motor vehicle or trailer (including a 'Semi-trailer') with a 'GVM' or 'ATM' less than 4.5 tonne provided that the load carrying capacity of each of the 'Axles' and the wheel and tyres fitted to each 'Axle' is at least 120% of the nominal load imposed on that 'Axle' with the vehicle or trailer at its 'GVM' or 'ATM';	

3.2 Functional Requirements: Normal Operation

Table 6: Functional requirements table (Normal conditions)

Reference	Title	Description	TPM
REQ1.1	Vehicle loading	Sustains full load acting on vehicle.	TPM101 – The full weight of SR8 is sustained by the suspension components.
REQ1.2	Vehicle moving and steering	Allows easy movement and steering of vehicle weight.	TPM102 – Vehicle can move and steer at variable speeds and loading.
REQ1.3	NVH	Isolates the occupant cell from the road to reduce NVH.	TPM103 – The NVH in the cabin is kept as low as possible while still achieving performance requirements
REQ1.4	Vibrations	Structurally withstands all road vibrations.	TPM104 – Wheels handle all road vibrations caused by tarred road travel (Track and public road)
REQ1.5	Braking	Withstands braking forces applied.	TPM105 – Braking forces exerted at variable speeds, temperatures and road condition are withstood in road and track conditions.

REQ1.6	Modularity	Accessible and replaceable within a required timeframe.	TPM106 – Suspension is modular in design and can be replaced within a short time frame with set tooling and torque settings.
REQ1.7	Versatility	Suspension is safe in all weather conditions.	TPM107 - Suspension can maintain functional requirements regardless of weather conditions.
REQ1.8	Maintenance	Suspension components have maintenance schedules and checking procedures for routine checks, pre track, post track and before all long road drives.	TPM108 – Suspension components have clear maintenance schedules and routines that align with use of SR8.
REQ1.9	Self-centering	Ensure that the wheels return to center if no load is applied to the steering wheel	TPM109 – Suspension geometry must exhibit a self-centering force on the wheels.

3.3 Performance Requirements: Teamwide Specifications

Table 7: Performance requirements (Teamwide specifications)

Reference	Title	Description	TPM
REQ2.1	Contact patch	Maintains adequate contact patch between tire and road in all driving conditions.	TPM201 – Tire camber angle must not exceed $\pm 4.5°$ of camber angle.
REQ2.2	Suspension mass	The total mass of the suspension should be kept at a minimum. The suspension components are counted as unsprung mass.	TPM202 – The total mass should be kept as low as possible while satisfying all load cases.
REQ2.3	System COG	Centre of gravity is critical to the handling of SR8.	TPM203 – The suspension components must contribute to the target COG height
REQ2.4	Mass distribution	The closer the suspension mass is located to the center, the easier the car will rotate.	TPM204 – The suspension components are kept as close to the center of the car as possible.
REQ2.5	Track performance	Suspension should optimize performance around the track. In accordance with the SR8 vision.	TPM205 – SR8 must be able to achieve the lap times as laid out in the SR8 vision document.
REQ2.6	Innovation	Suspension design should be innovative to generate interest in the project	TPM206 – Suspension design must include at least one innovative/high performance design.
REQ2.7	Sustainability	Suspension must be as sustainable as possible in terms of material use and manufacturing while still satisfying performance requirements.	TPM207 – Suspension is designed to be sustainable.

3.4 Interface Requirements: Project/System Interface

Table 8: Interface requirements (Project/System interfaces)

Reference	Title	Description	TPM
REQ3.1	Packaging	Interface between projects. Close collaboration with Packaging to enable other departments to interface with suspension and to determine the space in which suspension operate.	TPM301 - Packaging requirements for suspension travel and turning radius are met and allow full range of steering. Including required steering angle.
REQ3.2	Test Mule	Suspension to supply working product to test mule to enable testing of key data.	TPM302 - Test mule has accurate dimensions (Wheelbase and track) for effective testing and optimization. Suspension is supplied to test mule.
REQ3.3	Brakes	Interface is the connection between the suspension and the uprights of the brake system.	TPM303 – Suspension geometry is dependent on the geometry of the uprights, which are within the brakes project.
REQ3.4	Steering	Interface is the connection between the steering inputs and steering behaviour during suspension travel.	TPM304 - Wheelbase, track and key data is supplied for design of steering subsystem.
REQ3.5	Wheels	Interface is in the form of weight transfer and cornering g-force required to design and optimize the wheels subsystem.	TPM305 -Weight transfer, wheelbase, track, and cornering g-force is supplied. Allowing the design and optimisation of the wheels subsystem.
REQ3.6	Exterior	Exterior body panels must not interfere with the suspension at any point in the travel range.	TPM306 – Pass on full suspension travel information to the Exterior design project
REQ3.7	Occupant Cell	Occupant cell must not interfere with the suspension at any point in the travel range.	TPM307 – Pass on full suspension travel information to the occupant cell project
REQ3.8	Subframes	The suspension components mount to the chassis through the subframes	TPM308 – Inform the subframes project of the required mounting points for an ideal suspension geometry
REQ3.9	Active Aero	The suspension will need to be designed to accommodate the extra downforce exerted by the aero components when deployed	TPM309 – Obtain aero information from active aero project and design spring and damper system accordingly.

3.5 External Requirements

Reference	Title	Description	TPM
REQ5.1	Manufacturing Capabilities	Parts should be designed to allow efficient batch production especially	TPM501 – Generate CAM which enables multiple components

Commented [DR10]: Any requirements that come from our suppliers / partners?
i.e. How certain parts need to be designed to ensure best manufacturability

		if parts are machined or moulded	to be machined simultaneously
REQ5.2	Tool Life and Wear	Design features that are gentle on tooling, such as avoiding sharp internal corners, deep narrow slots and unnecessary tolerances	TPM502 – Approve CAD designs for CNC and lathe with the makerspace staff
REQ5.3	Standardise Components	Wherever possible use standardised sizes and parts (nuts, bolts, washers) that are readily available from suppliers	TPM503 – Metric sizing should be used across all suspension components
REQ5.4	Minimise Manufacturing steps	Parts should be designed to reduce the number of times the stock will need to be re-positioned and re-centred.	TPM504 – Approve CAD designs for CNC and lathe with the makerspace staff
REQ5.5	Joining Methods	Specify joining methods such as welding, press fitting and epoxy are within the campuses capabilities	TPM505 – Consult makerspace staff on universities joining capabilities and needed inductions